Spin representation via bosons

SHO $\quad [\hat{x}, \hat{p}] = i\hbar$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\,(a^+ + a)$$

$$\hat{p} = i\sqrt{\frac{\hbar m\omega}{2}}\,(a^+ - a)$$

$$[\hat{x}, \hat{p}] = \frac{i\hbar}{2}\,[a^+ + a,\ a^+ - a] = \frac{i\hbar}{2}\Big([\cancel{a^+, a^+}] - [a^+, a] + [a, a^+] - [\cancel{a, a}]\Big) = i\hbar \quad \text{when } [a, a^+] = 1$$

$$\left(\Rightarrow [a^+, a] = -1\right).$$

it follows that
$$\begin{cases} a^+ |n\rangle = \sqrt{n+1}\ |n+1\rangle \\ a\ |n\rangle = \sqrt{n}\ |n-1\rangle \end{cases}$$

then $a^+ a\ |n\rangle = a^+\left(\sqrt{n}\ |n-1\rangle\right) = \sqrt{n}\ \sqrt{n-1+1}\ |n-1+1\rangle =$
$= n\,|n\rangle$

$\Rightarrow \quad a^+ a = \hat{n} =$ number operator

Let us now introduce another pair of boson, b and b^+:

$[b, b^+] = 1 \quad$ but $\quad [b, a] = [b, a^+] = 0$

let
$$(1) \quad \begin{cases} S^z = \frac{\hbar}{2}\,(a^+ a - b^+ b) \\ S^+ = \hbar a^+ b \qquad S^x = \frac{1}{2}(S^+ + S^-) = \frac{\hbar}{2}\,(a^+ b + b^+ a) \\ S^- = \hbar b^+ a \qquad S^y = \frac{1}{2i}(S^+ - S^-) = \frac{\hbar}{2}\,(a^+ b - b^+ a) \end{cases}$$

Check identity ① $[S^+, S^-] = [S^x + iS^y,\ S^x - iS^y] = -i\,[S^x, S^y]$
$+ i\,[S^y, S^x] = -i\ i\hbar S^z + i\,(-i\hbar S^z) = 2\hbar S^z$

$$[a^+b, b^+a] = a^+ \underbrace{b\,b^+}_{1+b^+b} a - b^+ \underbrace{a\,a^+}_{1+a^+a} b =$$

$$= a^+a + a^+a\,b^+b - (b^+b + b^+b\,a^+a) = a^+a - b^+b$$

$$\Rightarrow \hbar^2 [a^+b, b^+a] = 2\hbar \cdot \frac{\hbar}{2}(a^+a - b^+b) = 2\hbar\, S^z.$$

② $[S^z, S^x] = i\hbar\, \underset{\overset{\parallel}{+1}}{\varepsilon^{zxy}} S^y$

$$\frac{\hbar^2}{4}[a^+a - b^+b,\ a^+b + b^+a] = \frac{\hbar^2}{4}\Big([a^+a, a^+b + b^+a]$$

$$- [b^+b, a^+b + b^+a]\Big) = \frac{\hbar^2}{4}\big(a^+b - b^+a - (b^+a - a^+b)\big)$$

because $[a^+a, a^+b] = a^+\underbrace{a\,a^+}_{1+a^+a} b - a^+b\,a^+a = a^+(1+a^+a)\,b - a^+a^+a\,b = a^+b$

$$[a^+a, b^+a] = a^+a\,b^+a - b^+a\,a^+a = b^+a^+aa - b^+(1+a^+a)a =$$

$$= -b^+a$$

$$\Rightarrow \frac{\hbar^2}{4}[a^+a - b^+b, a^+b + b^+a] = \frac{\hbar^2}{2}(a^+b - b^+a) =$$

$$= i\hbar \cdot \frac{\hbar}{2i}(a^+b - b^+a) = i\hbar\, S^y \quad !$$

③ Obviously $[S^z, S^y] = -i\hbar S^x$.

We also know that in the basis of angular momentum eigenstates $\hat{S}^2 = (S^x)^2 + (S^y)^2 + (S^z)^2 = \hbar^2 s(s+1)$

$$(S^x)^2 + (S^y)^2 = \left(\frac{S^+ + S^-}{2}\right)^2 + \left(\frac{S^+ - S^-}{2i}\right)^2 =$$

$$= \frac{1}{4}\big(S^+S^+ + S^-S^- + S^+S^- + S^-S^+ - S^+S^+ - S^-S^-$$

$$+ S^+S^- + S^-S^+\big) = \frac{1}{2}(S^+S^- + S^-S^+)$$

$$\Rightarrow \hat{S}^2 = \frac{\hbar^2}{4}\left(a^+a - b^+b\right)\left(a^+a - b^+b\right) + \frac{\hbar^2}{2}\left(a^+b\, b^+a + b^+a\, a^+b\right) = \frac{\hbar^2}{4}\Big\{ a^+a\, a^+a + b^+b\, b^+b$$

$$- a^+a\, b^+b - b^+b\, a^+a + 2\, a^+\underbrace{b\, b^+}_{1+b^+b}a + 2 b^+b\, \underbrace{a a^+}_{1+a^+a}\Big\}$$

$$= \frac{\hbar^2}{4}\Big\{ \overset{a^+aa^+a}{\cancel{a^+(1+a^+a)a}} + \overset{b^+bb^+b}{\cancel{b^+(1+b^+b)b}} + a^+a\, b^+b + b^+b\, a^+a + 2\left(a^+a + b^+b\right)\Big\} =$$

$$= \frac{\hbar^2}{4}\left(a^+a + b^+b\right)\left(a^+a + b^+b + 2\right) = \hbar^2 S(S+1)$$

Thus $\hat{S}^2 = \hbar^2 S(S+1)$ only if we <u>require</u>

$$\boxed{a^+a + b^+b = 2S} \qquad (2)$$

* Equations (1) and (2) form Schwinger boson representation of spin operator $\hat{S}$.

<u>Holstein – Primakoff representation</u> (H-P).

Let us "resolve" constraint (2) for b^+b:

$$b^+b = 2S - a^+a = \sqrt{2S - a^+a}\cdot\sqrt{2S - a^+a}$$

and require

$$b = b^+ = \sqrt{2S - a^+a} \quad \text{so that} \quad \boxed{a^+a \le 2S}$$

then $S^z = \frac{\hbar}{2}\left(a^+a - (2S - a^+a)\right) = \frac{\hbar}{2}\left(2a^+a - 2S\right) =$

$$= a^+a - S$$

$$(3)\quad \begin{cases} S^z = a^+a - S \\ S^+ = \hbar\, a^+\sqrt{2S - a^+a} \\ S^- = \hbar\sqrt{2S - a^+a}\, a \end{cases}$$

Could also do: $a^+a = 2S - b^+b = \sqrt{2S - b^+b}\cdot\sqrt{2S - b^+b}$

with $\underline{b^+b \le 2S}$

$$\Rightarrow\quad (3')\quad \begin{cases} S^z = \frac{\hbar}{2}\left(2S - b^+b - b^+b\right) = \hbar\left(S - b^+b\right). \\ S^+ = \hbar\sqrt{2S - b^+b}\, b \\ S^- = \hbar\, b^+\sqrt{2S - b^+b} \end{cases}$$

Can be checked that (3) and (3') satisfy spin commutation relations, for as long as $\begin{cases} a^+a \le 2S \\ b^+b \le 2S. \end{cases}$

H-P transformation is useful for <u>semi-classical</u> expansions about the ordered states, while Schwinger bosons are used for studying (quantum and thermal) disordered spin states.

Indeed, let us treat $\frac{b^+b}{2S}$ as a small parameter

$$\boxed{\frac{b^+b}{2S} \ll 1}$$

Then $S^z = \hbar\left(S - b^+b\right)$

$$S^+ = \hbar\sqrt{2S}\sqrt{1 - \frac{b^+b}{2S}}\; b \approx \hbar\sqrt{2S}\left(1 - \frac{b^+b}{4S} + \dots\right) b$$

$$= \hbar\sqrt{2S}\left(b - \frac{b^+b\,b}{4S} + \dots\right)$$

$$S^- = \hbar\, b^+ \sqrt{2S}\sqrt{1-\frac{b^+b}{2S}} \approx \hbar\sqrt{2S}\, b^+\left(1-\frac{b^+b}{4S}+\dots\right)$$

$$= \hbar\sqrt{2S}\left(b^+ - \frac{b^+b^+b}{4S}+\dots\right)$$

Notice that now we can <u>approximate</u>

$$S^+ = \hbar\sqrt{2S}\, b$$

$$S^- = \hbar\sqrt{2S}\, b^+$$

$$\Rightarrow [S^+, S^-] = \hbar^2(2S)\underbrace{[b, b^+]}_{1} = 2\hbar\cdot\hbar S$$

$$= 2\hbar\cdot\cancel{\hbar(S+b^+b)}\ \hbar S\left(1-\frac{b^+b}{S}\right) \approx 2\hbar\cdot\hbar S$$

i.e. the commutation relation is <u>okay</u> for as long as number of bosons <u>is small</u> : $\frac{b^+b}{S} \ll 1$.

Will see later that this has the meaning of "small" deviations from the classical ground state.

Many spins: Heisenberg ferromagnet

Spins from different sites commute $[S^a(r), S^b(r')] = 0$ for $r \neq r'$.

$$H = -\frac{1}{2} J \sum_{n,g} \vec{S}_n \cdot \vec{S}_{n+g}$$

For any pair of spins: $\vec{S}_n \cdot \vec{S}_{n+g} = S_n^z \cdot S_{n+g}^z + \frac{1}{2}(S_n^+ S_{n+g}^- + S_n^- S_{n+g}^+)$

$$= (S - b_n^+ b_n)(S - b_{n+g}^+ b_{n+g}) + \frac{1}{2}(2S)(b_n b_{n+g}^+ + b_n^+ b_{n+g}) =$$

$$= S^2 - S(b_n^+ b_n + b_{n+g}^+ b_{n+g} - b_n^+ b_{n+g} - b_{n+g}^+ b_n)$$
$$+ b_n^+ b_n \cdot b_{n+g}^+ b_{n+g}$$

$$H = -\frac{1}{2} J \Big\{ \underbrace{S^2 \cdot zN}_{\substack{\text{classical energy} \\ \text{(every spin has } z \text{ neighbors)}}} - S \sum_{n,g} (b_n^+ b_n + b_{n+g}^+ b_{n+g} - b_n^+ b_{n+g} - b_{n+g}^+ b_n)$$
$$+ \sum_{ng} \underbrace{b_n^+ b_n \cdot b_{n+g}^+ b_{n+g}}_{\text{Interaction between spin waves}} \Big\}$$

Observe that last term does not have factor (S) in front of the sum $\Rightarrow$ it gives $\frac{1}{S}$ contribution in comparison with the second sum

$\Rightarrow$ small for large spins, $\underline{S \gg 1}$

$$H = H^{(0)} + H^{(2)} + H^{(4)}$$

where
$$\left.\begin{aligned} H^{(0)} &\sim S^2 \\ H^{(2)} &\sim S \\ H^{(4)} &\sim O(1) \end{aligned}\right\} \begin{aligned} &\rightarrow \text{linear spin wave approximation} \\ &\rightarrow \text{neglect } O(1) \text{ contribution!} \end{aligned}$$

$$H \approx H^{(0)} + H^{(2)}.$$

$$H^{(2)} = \frac{1}{2} JS \sum_{ng} \left(2 b_n^+ b_n - b_n^+ b_{n+g} - b_{n+g}^+ b_n\right)$$

Fourier transform to momentum space:

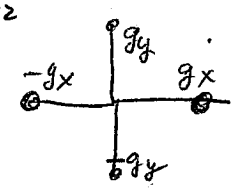
$$b_n = \frac{1}{\sqrt{N}} \sum_k e^{i\bar{k}\bar{R}_n} b_k$$

$$b_n^+ = \frac{1}{\sqrt{N}} \sum_k e^{-i\bar{k}\bar{R}_n} b_k^+$$

$$H^{(2)} = \frac{1}{2} JS \frac{1}{N} \sum_{k_1 k_2} \sum_{ng} \Big(2 b_{k_1}^+ b_{k_2} e^{-i\bar{R}_n(\bar{k}_1-\bar{k}_2)}$$

$$- e^{-i\bar{R}_n(\bar{k}_1-\bar{k}_2)} e^{+i\bar{g}\bar{k}_1} b_{k_1}^+ b_{k_2} - e^{-i\bar{R}_n(\bar{k}_1-\bar{k}_2)} e^{-i\bar{g}\bar{k}_1} b_{k_1}^+ b_{k_2}\Big)$$

$$= \frac{1}{2} JS \frac{1}{N} \sum_{k_1 k_2} \underbrace{\sum_{R_n} e^{-i\bar{R}_n(\bar{k}_1-\bar{k}_2)}}_{N\Delta_{\bar{k}_1,\bar{k}_2} \Rightarrow \bar{k}_1=\bar{k}_2} \sum_g b_{k_1}^+ b_{k_2} (2 - 2\cos\bar{g}\cdot\bar{k}_1)$$



$$= JS \sum_k \varepsilon_k b_k^+ b_k$$

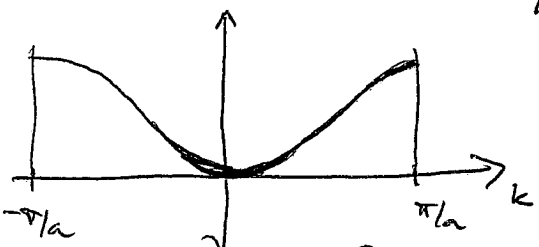
where $\varepsilon_k = \sum_g (1 - \cos\bar{k}\cdot\bar{g}) = 2(1-\cos k_x a) + 2(1-\cos k_y a) + 2(1-\cos k_z a) =$

$$= 6 - 2(\cos k_x a + \cos k_y a + \cos k_z a)$$

Spin wave dispersion : $JS\,\varepsilon(k) \to SJa^2 \vec{k}^2$ for small momenta

<u>min</u> $\varepsilon(0) = 0$

<u>max</u> $\varepsilon(\vec{\pi}) = 12$



$\cos ka \approx 1 - \frac{1}{2}k^2a^2$ for $k \to 0$

$$\Rightarrow \varepsilon(k) \approx 6 - 2\left(3 - \frac{a^2}{2}(k_x^2+k_y^2+k_z^2)\right) = a^2\vec{k}^2 + \ldots$$

quadratic dispersion near the origin!

Thermodynamic limit

$$\frac{1}{N}\sum_k \cdots = v_0 \int_{BZ} \frac{d^D k}{(2\pi)^D} \cdots$$

volume of the lattice unit cell

$$N \cdot v_0 = V = \text{volume}$$

$$H^{(2)} = \sum_k JS\varepsilon(k)\, b_k^+ b_k \quad = \text{ideal gas of bosons}$$

Number of bosons with momentum $\vec{k}$ at temperature T:

$$n_k = \langle b_k^+ b_k \rangle = \frac{1}{e^{\frac{JS\varepsilon(k)}{k_B T}} - 1} \quad \text{Bose-Einstein distribution}$$

average over thermal ensemble

$$\langle H^{(2)} \rangle = E^{(2)} = \sum_k JS\varepsilon(k) \langle b_k^+ b_k \rangle$$

$$= \underbrace{N_0 v_0}_{\text{Volume}} \int \frac{d\vec{k}}{(2\pi)^3} \frac{JS\varepsilon(k)}{\exp\left(\frac{JS\varepsilon(k)}{k_B T}\right) - 1}$$

But we expect bosons with small $\vec{k}$ (lowest energy) to contribute most $\Rightarrow SJ\varepsilon(k) \approx D\vec{k}^2$ ($D = SJa^2 =$ spin stiffness)

$$E^{(2)} = V \int \frac{d\vec{k}}{(2\pi)^3} \frac{D\vec{k}^2}{e^{\frac{D\vec{k}^2}{k_B T}} - 1} =$$

$$= V \cdot \frac{4\pi}{(2\pi)^3} \int_0^\infty dk \cdot k^2 \frac{Dk^2}{e^{\frac{Dk^2}{k_B T}} - 1} =$$

$$= \left(k = \sqrt{\frac{k_B T}{D}}\, x\right) = \frac{V}{2\pi^2} \left(\frac{k_B T}{D}\right)^{\frac{1}{2}\cdot 5} \int_0^\infty \frac{dx\, x^4}{e^x - 1}$$

$$\Rightarrow E^{(2)} \sim T^{5/2}$$

$$C_V = \frac{1}{V}\left(\frac{\partial E}{\partial T}\right)_V \sim T^{3/2}$$

Total number of bosons

$$N = \sum_k n_k = V \int \frac{d\vec{k}}{(2\pi)^3} \frac{1}{e^{\frac{Dk^2}{k_B T}} - 1} =$$

$$\simeq \frac{V}{2\pi^2} \int_0^\infty dk \cdot k^2 \frac{1}{\exp\left(\frac{Dk^2}{k_B T}\right) - 1}$$

$$= \frac{V}{2\pi^2} \left(\frac{k_B T}{D}\right)^{3/2} \int_0^\infty \frac{dx\, x^2}{e^x - 1}$$

Every boson reduces S_{tot} by 1

$\Rightarrow$ on-site ~~magnetic~~ spin $\langle S^z(r) \rangle = S - \langle b_r^+ b_r \rangle$

$$= S - \frac{1}{N} \sum_k n_k = S - (\#) \cdot T^{3/2}$$

Bloch's law.